



DEALER BEST PRACTICE SERIES

Unavailability Distribution – Effectively Manage the Unavailability of your Fleet

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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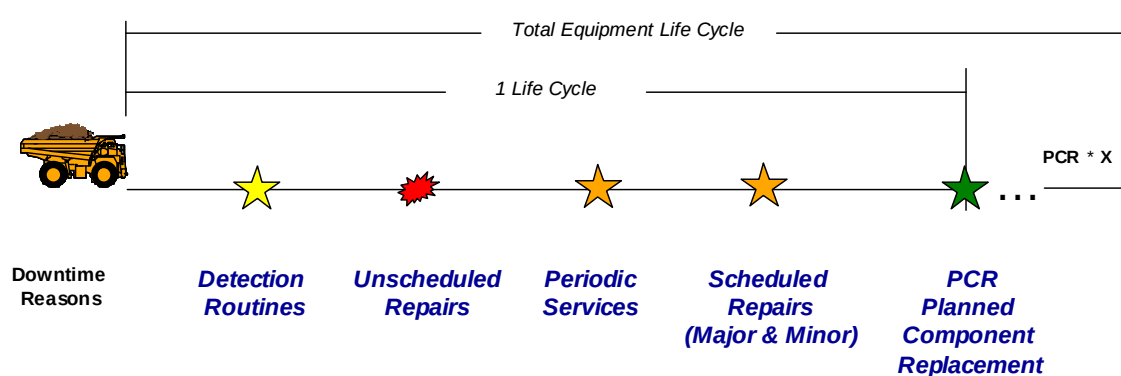
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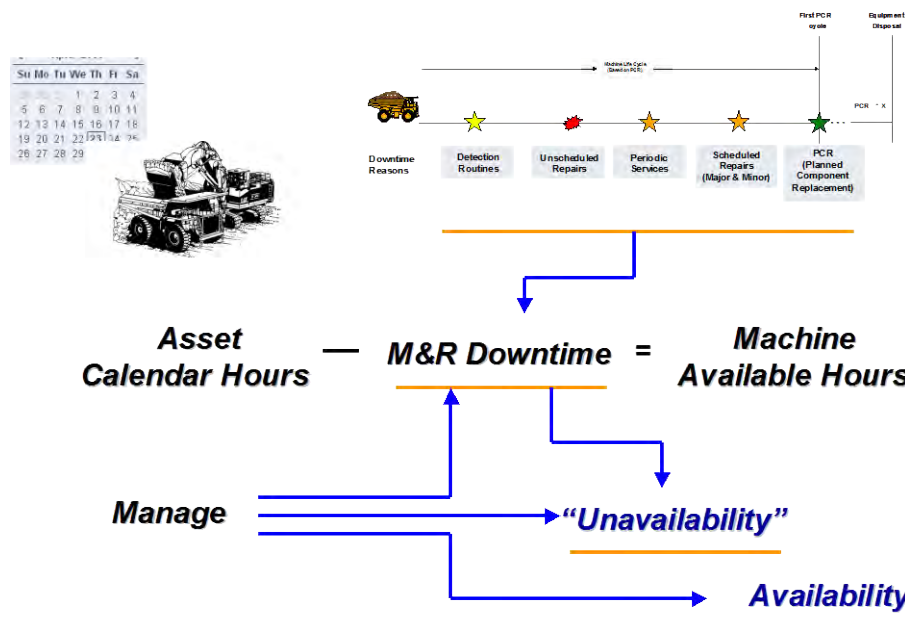
1.0 Introduction

Availability is the most common metric currently use in the mining industry to measure equipment fleet performance. The main reason is that the production plan is based on a specific amount of equipment hours. The mine production department determines the required hours for the production and ancillary equipment, defining the utilization and availability targets.

The difference between the calendar hours and the availability target is the unavailable time. The maintenance department has to use this time efficiently to perform all maintenance and repair events needed to maintain the equipment running.



This diagram illustrates the various types of maintenance events that will be considered in the maintenance plan. These events take time away from the asset being utilized. Our objective as the maintenance organization is to manage the unavailable time effectively to meet and support the production plan requirements.



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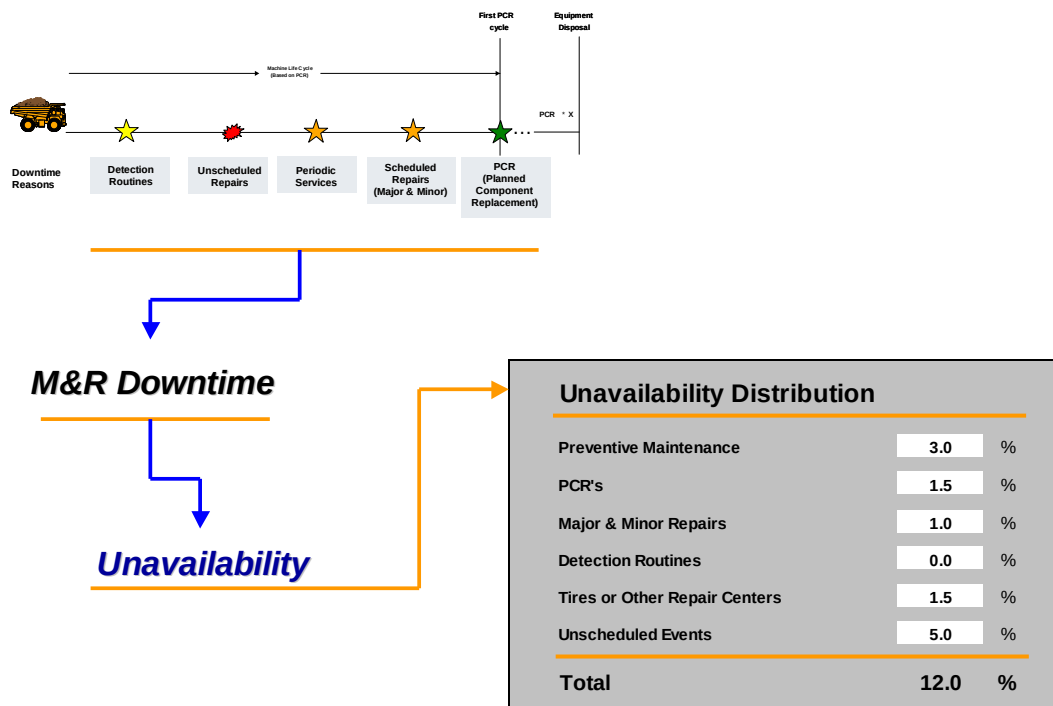
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2.0 Best Practice Description

The Maintenance Organization as part of the production chain of the mine takes the availability target as one of their primary performance indicator. The availability not only defines the needed hours to support the production plan, but also the amount of hours that can be dedicated to Maintenance & Repair (M&R) activities. The goal is to effectively distribute these available hours for M&R activities to execute the monthly scheduled and unscheduled events.

As another way to understand the concept, the maintenance personnel manage the time of the equipment when they are “down” and the production department does the same with the “up” time.

There are some maintenance operations that daily track time used for M&R activities and this compared with the available time for maintenance. In that way they are able to make adjustments to prevent availability shortfalls at the end of the period. It is the first step to start getting control of the M&R time. This basic approach can be improved and made more effective by separating the available time into the various event categories, and start tracking the time for each category of events. We call this unavailable time categorization the, “Unavailability Distribution”.



The diagram above illustrates the sequence of transforming each of the M&R events into downtime, unavailability and finally represented in the unavailability distribution.

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Building the Unavailability Distribution

We will use a calculation example to explain the steps to follow when building the unavailability distribution.

1. - The first step is to acquire the equipment fleet characteristics, the availability, and utilization goals.

Case-scenario for our calculations:

Equipment Type	793 OHT
Number of Machines	30
Physical Availability Goal	87%
Utilization of Asset	81%
Month	January

2. – With the information provided we analyze and determine the hours distribution in the period.

Calendar hours = Number of machines x days in the period x 24 (h)

Available hours (Physical) = Calendar hours x Physical availability (h)

Operating hours = Calendar hours x Utilization of Asset (h).

It is convenient to remember at this point the formula for physical availability:

$$\text{Physical Availability} = \frac{\text{Total Calendar Hours} - \text{Maintenance \& Repair Hours}}{\text{Total Calendar Hours}} \times 100\%$$

Note: care must be taken when dealing with utilization of asset or utilization of availability. In some countries when you refer to utilization it is always utilization of availability.

Calculations for our case scenario

Calendar hours = $30 \times 31 \times 24 = \mathbf{22,320 \text{ (h)}}$

Available hours = $22,320 \times 87\% = \mathbf{19,418 \text{ (h)}}$

Operating hours = $22,320 \times 81\% = \mathbf{18,079 \text{ (h)}}$

Now the difference between the calendar hours and the available hours is the amount of hours available to do all maintenance and repair activities.

Hours for M&R = $22,320 - 19,418 = \mathbf{2,902 \text{ (h)}}$

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And the stand by hours, or hours that the equipment is available but not being utilized is

Stand By hours = Available hours – Operating hours = **1,339 (h)**

It is important to know the stand by hours to understand the window of opportunities that the production department may provide the maintenance department to do short type of maintenance activities, such as inspections.

3. – Building the Unavailability Distribution

We will use the following template to present our results

Unavailability Distribution	Percentage	Hours
Preventive Maintenance		
Planned Component Replacement (PCR)		
Major / Minor Scheduled Repairs		
Detection Routines		
Tires		
Other Repair Centers		
Unscheduled Repairs		
TOTAL UNAVAILABILITY	0.00%	0
ESTIMATED AVAILABILITY	100.00%	
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
AVAILABILITY GOAL	87.00%	19418

For each of the 7 categories, listed above, we have to determine the percentage and hours that we will dedicate to them.

In the template we have already recorded information and calculations from Step 2. With this information the planner can start his or her projection analysis for each of the event categories.

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Preventive Maintenance (PM) Services

The PM services in this scenario will follow the 8 Step strategy with an interval of 250 hours and a target duration for each step of 10 hours.

First, determine the number of PM services to execute in the month:

Number of PMs to execute = Operating hours (Fleet) / PM Interval

PMs to execute = 18,079 h / 250 h = **72.3**

The total duration that we will have to dedicate to PM this month will be:

PM total duration = Number of PMs x PM duration
= 72.3 x 10 h = **723 (h)**

Now the unavailability due to the PM services will be:

PM unavailability = 1-(Calendar hours – PM total duration time)/Calendar hours = (%)

Or a simpler formula:

PM unavailability = PM time total duration time / Calendar hours = (%)

PM unavailability = 723 (h) / 22,320 (h) = **3.24 (%)**

These are our first results to enter into our distribution table:

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)		
Major / Minor Scheduled Repairs		
Detection Routines		
Tires		
Other Repair Centers		
Unscheduled Repairs		
TOTAL UNAVAILABILITY	3.24%	723
ESTIMATED AVAILABILITY	96.76%	21597
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
AVAILABILITY GOAL	87.00%	19418

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Planned Component Replacement (PCR)

For the PCR's, the planner builds the Long and Medium Term Plan confirming the machines selected to be scheduled for PCR's. After the pre-PCR inspection the planner builds the final repair plan(s) defining the duration and resources required. Please refer to BP 1110-1.03-1260.

In our case study we will use the following information.

Three machines have been selected to perform the PCR. The duration decided by Planning is:

Truck 1	6.5 days
Truck 2	10.0 days
Truck 3	4.75 days

The total duration in days is: **21.25**

And expressed in hours (machine downtime): 21.25 days x 24 hrs = **510 h**

PCR's unavailability = Total PCR duration (h) / Calendar hours (h) [%]

PCR's unavailability = 510 (h) / 22,320 (h) (%) = **2.28 (%)**

And the unavailability distribution will look like:

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)	2.28%	510
Major / Minor Scheduled Repairs		
Detection Routines		
Tires		
Other Repair Centers		
Unscheduled Repairs		
TOTAL UNAVAILABILITY	5.52%	1233
ESTIMATED AVAILABILITY	94.48%	21087
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
REMAINING HOURS FOR M&R EVENTS		1668
AVAILABILITY GOAL	87.00%	19418

So far we have committed 1233 hours or 5.52% of availability for PM services and PCR's. We still have 1668 hours remaining to plan the rest of the maintenance and repair activities without compromising the availability target.

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Detection Routines

In the case of the detection or condition monitoring, these type of activities may or may not take away available hours of the equipment. Most of these activities are done in parallel with another maintenance activity, such as PM services, and others are completed in a window of opportunity that the operation characteristic may offer. In this last case the use of operational window of opportunity must be agreed with the production department.

In our case study we will use the following scenario.

Field inspections duration = 17 min
 Frequency of field inspections = 3 days
 Field inspections to perform in
 Window of opportunity = 70%
 Inspection program compliance target = 90%

Pre-PM services inspections duration = 30 min
 Pre-PM inspections performed in
 Window of opportunity = 0%
 Pre-PM inspections compliance target = 95%

For the Field Inspections:

Inspections to perform in the month = 30 (machines) x 31 (days)/ 3 (frequency) x 90% (comp. Target)
 Inspections to perform in the month = **279**

Total duration of Field Inspections = 279 (number of Insp.) x 17/60 (time in h format)
 Total duration of Field Inspections = **79.05 h**

Total duration that will affect the availability = 79.05 x 30% (Inspections to perform outside a window of opportunity)
 Total duration affecting availability = **24 h**

For the Pre-PM Inspections:

Pre-PM inspections to perform = PM services to perform = **72.3**
 Target based on compliance goal = 72.3 x 95% = **69**

Total duration of Pre-PM inspections = 69 (Pre-PM Inspections) X 0.5h (Inspection duration)
 = **34.35 h**

For the Pre-PCR Inspections:

At this stage we have to also consider the Pre-PCR inspections that will have to be performed in the month. The duration of each Pre-PCR inspections will have to be determine based on the performance of the machine and in the amount of unidentified problems that the machine may be

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experimenting. Some operations schedule the Pre-PCR inspections for a shift, but this is not a mandatory rule. It can be any time that must be determined by the Fleet Analyst and the Planner. In this case for our example we will consider a duration of 8 hours for each Pre-PCR inspection.

In our case scenario we have 3 PCR scheduled for the month, so we will have same number of Pre-PCR inspections.

Total duration for the Pre-PCR inspections = **24 h**

The total downtime to consider for the detection routines = 24 h + 34.35 h + 24 h
= **82.07 h**

Unavailability to consider for the Detection Routines = **0.37 %**

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)	2.28%	510
Major / Minor Scheduled Repairs		
Detection Routines	0.37%	82
Tires		
Other Repair Centers		
Unscheduled Repairs		
TOTAL UNAVAILABILITY	5.89%	1315
ESTIMATED AVAILABILITY	94.11%	21005
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
REMAINING HOURS FOR M&R EVENTS		1586
AVAILABILITY GOAL	87.00%	19418

Major and / or Minor Scheduled Repairs

The information for our case scenario indicates:

M&M scheduled repairs to be performed every 500 hours with duration of 4 hours. This is a decision of the planning department that is normally based on the Backlog workload. We will discuss this subject (defining the BL execution targets) in another Best Practice.

The information provided also call for 2 repairs to be done because of the critical nature of the repairs needed. These 2 two repairs are planned for a duration of one shift of 12 h each.

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Duration for the M&M by frequency = (Operating hours / Frequency) x Duration [h]

Duration M&M freq = (18,079 / 500) x 4 = **144.6 h**

Duration for the repairs by exception = Repair Duration 1 + Repair Duration 2 [h]

Duration for the repairs by exception = **24 h**

Total Duration for the M & M scheduled repairs = 144.6+24 = **168.6 h**

Unavailability due to M&M repairs = Total Duration M&M / Calendar hours [%]

Unavailability due to M&M repairs = 168.6 / 22,320 = **0.76 %**

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)	2.28%	510
Major / Minor Scheduled Repairs	0.76%	169
Detection Routines	0.37%	82
Tires		
Other Repair Centers		
Unscheduled Repairs		
TOTAL UNAVAILABILITY	6.65%	1484
ESTIMATED AVAILABILITY	93.35%	20836
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
REMAINING HOURS FOR M&R EVENTS		1418
AVAILABILITY GOAL	87.00%	19418

This is how the distribution looks like after including the M&M scheduled repairs.

Tires

Tires are a sensitive issue for the success of the mining operation in regards to cost and downtime allocation. A dedicated department that works in parallel with the maintenance and repair shop normally takes the tire maintenance and repair activities. Coordination between the two entities is vital. The planning department of the equipment has to work together with the tire shop to determine and understand the time that will be needed for tire maintenance.

In our scenario we have the following situation:

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Daily tire inspection = 15 minutes per machine
 Program compliance target = 90%
 Tire checks within Window of Opportunity = 80%

Tire shop monthly activities = 255 hours

Inspections to perform monthly = Number of Machines x 31 x Compliance Target
 = 30 X 31X 90% = **837**

Inspection time to affect availability = 837 x 20% x 15 / 60
 = **41.85 h**

Total unavailable hours due to Tires = Tire checks + in shop time = 41.85 + 255 h
 = **296.85 h**

Unavailability due to Tires = Tires unavailable hours / Calendar hours = 296.85 / 22,320
 = **1.33%**

Other Repair Centers

Welding repair shop, radio communications, special attachments are some examples of other repair centers that at some time may take away time from the machines available hours. In our example we will use the requirement of 1 body exchange that will be scheduled for 2 shifts of 12 hours.

Total time to use will be = **24 h**

The unavailability due to this work will be = **0.11 %**

The unavailability distribution including these last two categories will look like,

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)	2.28%	510
Major / Minor Scheduled Repairs	0.76%	169
Detection Routines	0.37%	82
Tires	1.33%	297
Other Repair Centers	0.11%	24
Unscheduled Repairs		
TOTAL UNAVAILABILITY	8.09%	1805
ESTIMATED AVAILABILITY	91.91%	20515
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
REMAINING HOURS FOR M&R EVENTS		1097
AVAILABILITY GOAL	87.00%	19418

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So far we defined and included in the Unavailability Distribution planned events with the exception of the tire estimated time in where there is an amount of hours / repairs that will have to be done under an unscheduled scenario.

The last item to consider is the time provision to be made for the unscheduled repairs.

Unscheduled Repairs

To determine the amount of hours to include in this unavailability distribution we have to analyze the repair history, and the results of the past 2 – 3 months. We cannot predict the future, but we can make our best estimate based on past performance.

In our case study we will work with a fleet that has been down because of unscheduled events for **1545 hours**. This is the average of the last 3 months.

The unavailability due to the unscheduled repairs will be,

$$\text{Unscheduled downtime} / \text{Calendar hours} = 1545 / 22,320 [\%] = \mathbf{6.92 \%}$$

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)	2.28%	510
Major / Minor Scheduled Repairs	0.76%	169
Detection Routines	0.37%	82
Tires	1.33%	297
Other Repair Centers	0.11%	24
Unscheduled Repairs	6.92%	1545
TOTAL UNAVAILABILITY	15.01%	3350
ESTIMATED AVAILABILITY	84.99%	18970
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
REMAINING HOURS FOR M&R EVENTS		-448
AVAILABILITY GOAL	87.00%	19418

At this time, after including our estimate for the unscheduled repairs, we end up in a situation in which the forecasted unavailability is greater than the allowed (15.01% > 13%) to meet the availability goal.

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The first action to take is to try to adjust the allocated M&R time to reduce the total unavailable hours. In this scenario we could review the following options:

- Increase the Backlog execution in PM services and reduce the M&M duration of the repairs to be done by frequency.
- If we are in a very demanding situation in regards to production we could also stop one of the PCR work and postpone it for the next month or scheduling period. This decision has to be based on an accurate analysis of the conditions of the component.
- Increase the amount of detection routines that are performed within a window of opportunity.
- Try to reduce the PM service duration by working on the efficiency of its execution. This action is not an immediate action that can be taken; it is a medium term effort.
- Other actions may be taken based on the particular situation that we may be confronted with.

Unavailability Distribution	Percentage	Hours
Preventive Maintenance	3.24%	723
Planned Component Replacement (PCR)	1.21%	270
Major / Minor Scheduled Repairs	0.43%	96
Detection Routines	0.33%	74
Tires	1.33%	297
Other Repair Centers	0.11%	24
Unscheduled Repairs	6.92%	1545
TOTAL UNAVAILABILITY	13.57%	3029
ESTIMATED AVAILABILITY	86.43%	19291
TOTAL HOURS TO EXECUTE M&R EVENTS		2902
REMAINING HOURS FOR M&R EVENTS		-128
AVAILABILITY GOAL	87.00%	19418

The above chart is the representation where we applied some of the adjustment ideas. We eliminated 1 PCR, reduced the M&M time by reducing the allocated time for M&M by frequency with the agreement to execute more backlogs in parallel with the PM services, and increase the target for using the Windows of Opportunity for inspections. Reduction of the estimated time for the unscheduled repairs is not suggested; it would be just a mathematical adjustment. What we can do with the unscheduled repairs is to ensure that the unscheduled hours doesn't increase and

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to set effective machine performance targets for the scheduled repairs to be performed. This will give better control of the future unscheduled repairs reduction.

At the end when maintenance planning cannot adjust any further without getting in an uncontrollable risk it comes a time to present the situation with the production department of the customer. The production department can adjust their operation plan and also try to increase the utilization of the available machines.

3.0 Implementation Steps

The implementation of this concept is relatively simple. This concept and procedure has to be the responsibility of the on-site planner; it is a strategic task.

The steps to follow in the implementation are as follows:

- ✓ Evaluate your planning process. Determine with the “As Is” situation if your current process. Consider the use of the proposed procedure in this BP.
- ✓ Design a spreadsheet with the various categories to be included. Create your Unavailability Distribution template. You can use the format suggested in this BP.
- ✓ Determine the equipment fleet operating parameters (utilization, availability target, ...).
- ✓ Investigate and gather all needed information.
- ✓ Populate your Unavailability Distribution.
- ✓ Present the concept and tool to your organization. This step is very important to get the “buy in” of your organization. The success of this procedure depends on the collective work of the on-site organization.
- ✓ Incorporate the Unavailability Distribution in all your planning meetings.
- ✓ Link the Long, Medium and Short-Term plans to the Unavailability Distribution process.
- ✓ Follow up. Closely monitor the used of the remaining hours for performing the maintenance and repair activities.
- ✓ Revise and update as necessary.

4.0 Benefits

The use of the Unavailability Distribution concept described in this BP will give your maintenance organization a better understanding of their particular roles and responsibilities in meeting the expected fleet performance goals. It will also provide an effective way to work with specific goals linked to each functional area of your operation.

For Planning, who is responsible of building and leading this effort, the benefits are:

- ✓ Clear goals and targets for following up and controlling the execution of the scheduled tasks.
- ✓ Better management of the adjustments that may be necessary.
- ✓ Gain control.
- ✓ Moving from expecting “good” things to occur to a controlled and decision making situation

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For the Work Shop crews (PM services, unscheduled, PCR's, etc.)

- ✓ Clear set of goals to guide and meet the required fleet available time.
- ✓ Better information to make better decision (When and when not to extend or do extra repairs).
- ✓ Clear identification of opportunities.
- ✓ Better management of maintenance activities.
- ✓ Integration of the efforts with the rest of the areas of the maintenance organization.

For the Operation Management

- ✓ Improve the control and management of the operation with effective and updated information.
- ✓ Make the appropriate adjustments to the right areas.
- ✓ Manage the unavailable time by clear targets.
- ✓ Better fleet equipment management.

For the Production Department

- ✓ Better understanding of the tasks and targets of the maintenance department.
- ✓ Better integration. Increase the team effort.
- ✓ Better planning of the production plan with better information.

It is convenient to mention at this stage that this procedure has been implemented in the various operations as part of the M&R process implementation. I would like to highlight the operations of:

- Trakindo, Indonesia in Batu Hijau, Newmont operation.
- TIL, India in SEB, Tata Steel operation.

The results are very encouraging and benefits in better control and management of the equipment fleet are very solid.

Trakindo at Batu Hijau has extended this procedure to incorporate additional information to the tool that can even increase its effectiveness.

5.0 Resources Required

The implementation of this BP will not require any extra manpower or infrastructure; however, depending on your particular situation it will be necessary to redefine the function, role, and responsibility of the planner. Special attention must be dedicated to the planner who gets primarily focused in building and structuring the Short-Term Plan. Building the Short-Plan is a scheduling function. In this situation we recommend separating the functions of planning and scheduling, and redistributing both functions.

The procedure presented in this BP is a responsibility of the planner. It is part of building strategic guidelines for the organization to follow.

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6.0 Supporting Attachments / References

No attachments.

7.0 Related Best Practices

- **BP 1110-1.03-1260**, “Moving the Maintenance Strategy into an actionable plan”
- **BP 1007-2.0-1104** “Strategic Planning & Scheduling BP”

8.0 Acknowledgements

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